Challenge 39: Alteration of cavity field coherences

Steady-state cavity coherences $\langle n'|\hat\rho_{c,\mathrm{ss}}|n\rangle$, with a squeezed-input bonus

Companion implementation: `challenges/39/answer.py` · standalone QuTiP script: `challenges/39/qutip_check.py`

1 Statement

A three-level atom with ground states $|b\rangle, |d\rangle$ and excited state $|e\rangle$ sits in a lossless cavity resonant with the bright transition $|b\rangle \leftrightarrow |e\rangle$. With $\hbar = 1$,

$$\hat H = \frac{g}{2}\left(|b\rangle\langle e|\hat a^\dagger + |e\rangle\langle b|\hat a\right), \qquad \hat J = \sqrt{\gamma}\,|d\rangle\langle e|, \qquad \dot{\hat\rho} = -\mathrm{i}[\hat H,\hat\rho] + \hat J\hat\rho\hat J^\dagger - \tfrac12\{\hat J^\dagger\hat J,\hat\rho\}, \tag{1}$$

and $\hat\rho_0 = |b\rangle\langle b| \otimes |\alpha\rangle\langle\alpha|$. We want $\langle n'|\hat\rho_{c,\mathrm{ss}}|n\rangle$.

Careful: $\mathcal{L}\hat\rho = 0$ does not determine the answer

The Liouvillian of Eq. (1) has a massively degenerate kernel: every $|d\rangle\langle d| \otimes \hat\sigma$ with arbitrary cavity state $\hat\sigma$ is stationary, and so is $|b,0\rangle\langle b,0|$. Setting the left-hand side to zero therefore gives a family of solutions, and the physical steady state is selected by the *initial condition.* The clean way to impose "LHS = 0" is to integrate Eq. (1) over all time,

$$\int_0^\infty \mathrm{d}t\,\dot{\hat\rho} = \hat\rho(\infty) - \hat\rho(0), \tag{2}$$

which turns the equations of motion into an *algebraic* linear system for the time integrals $\hat M_{xy} = \int_0^\infty \hat\rho_{xy}(t)\,\mathrm{d}t$. That is the derivation below. (In the same vein, `qutip.steadystate()` returns an arbitrary kernel element here and must not be used — the numerics in Sec. 4 propagate to long times instead.)

2 Derivation

2.1 Block equations of motion

Write $\hat\rho = \sum_{x,y\in\{b,e,d\}} |x\rangle\langle y| \otimes \hat\rho_{xy}$, where each $\hat\rho_{xy}$ is an operator on the cavity. Substituting into Eq. (1) and collecting blocks,

$$\dot{\hat\rho}_{bb} = -\tfrac{\mathrm{i}g}{2}\left(\hat a^\dagger\hat\rho_{eb} - \hat\rho_{be}\hat a\right), \tag{3}$$

$$\dot{\hat\rho}_{be} = -\tfrac{\mathrm{i}g}{2}\left(\hat a^\dagger\hat\rho_{ee} - \hat\rho_{bb}\hat a^\dagger\right) - \tfrac{\gamma}{2}\hat\rho_{be}, \tag{4}$$

$$\dot{\hat\rho}_{eb} = -\tfrac{\mathrm{i}g}{2}\left(\hat a\hat\rho_{bb} - \hat\rho_{ee}\hat a\right) - \tfrac{\gamma}{2}\hat\rho_{eb}, \tag{5}$$

$$\dot{\hat\rho}_{ee} = -\tfrac{\mathrm{i}g}{2}\left(\hat a\hat\rho_{be} - \hat\rho_{eb}\hat a^\dagger\right) - \gamma\hat\rho_{ee}, \tag{6}$$

$$\dot{\hat\rho}_{dd} = \gamma\hat\rho_{ee}. \tag{7}$$

The blocks $\hat\rho_{bd}, \hat\rho_{ed}$ and their adjoints start at zero and are never sourced ($\hat J$ produces only $|d\rangle\langle d|$ terms, and $\hat H$ does not touch $|d\rangle$), so they stay zero: **no atomic coherence with the dark state is ever generated**, and the cavity state factorises as $\hat\rho_c = \hat\rho_{bb} + \hat\rho_{ee} + \hat\rho_{dd}$.

2.2 Integrating over all time

As $t \to \infty$ every excitation ends in $|d\rangle$ except the piece that started in the dark corner $|b,0\rangle$ (for which $\hat{a}|b,0\rangle = 0$, so it does not evolve). Hence

$$\hat{\rho}_{bb}(\infty) = \rho^0_{00}|0\rangle\langle 0|, \qquad \hat{\rho}_{be}(\infty) = \hat{\rho}_{ee}(\infty) = 0, \qquad \hat{\rho}^{\text{ss}}_c = \rho^0_{00}|0\rangle\langle 0| + \hat{\rho}_{dd}(\infty), \tag{8}$$

with $\rho^0_{n'n} = \langle n'|\hat{\rho}_c(0)|n\rangle$. Integrating Eqs. (3)–(6) using (2) and (8) gives four algebraic equations for $\hat{M}_{bb}, \hat{M}_{be}, \hat{M}_{eb}, \hat{M}_{ee}$, and (7) gives the answer directly:

$$\hat{\rho}_{dd}(\infty) = \gamma \hat{M}_{ee}. \tag{9}$$

2.3 Solving, manifold by manifold

The operators $\hat{a}, \hat{a}^\dagger$ shift Fock indices by one, so the system closes on each pair (n, n') with $n, n' \geq 1$. Writing

$$\hat{M}_{bb} \supset u|n\rangle\langle n'|, \quad \hat{M}_{ee} \supset v|n-1\rangle\langle n'-1|, \quad \hat{M}_{be} \supset w|n\rangle\langle n'-1|, \quad \hat{M}_{eb} \supset x|n-1\rangle\langle n'|, \tag{10}$$

and abbreviating $\Omega_n = \frac{g}{2}\sqrt{n}$ (the matrix element $\langle e, n-1|\hat{H}|b, n\rangle$), Eqs. (6), (4), (5), (3) become

$$\gamma v = -\mathrm{i}\left(\Omega_n w - \Omega_{n'} x\right), \qquad \tfrac{\gamma}{2} w = -\mathrm{i}\left(\Omega_n v - \Omega_{n'} u\right), \tag{11}$$

$$\tfrac{\gamma}{2} x = -\mathrm{i}\left(\Omega_n u - \Omega_{n'} v\right), \qquad -\rho^0_{nn'} = -\mathrm{i}\left(\Omega_n x - \Omega_{n'} w\right). \tag{12}$$

Eliminating w, x from the two right-hand equations and inserting into the two left-hand ones gives, with $S = \Omega_n^2 + \Omega_{n'}^2$ and $P = \Omega_n \Omega_{n'}$,

$$v\left(\gamma^2 + 2S\right) = 4Pu, \qquad \rho^0_{nn'} = \frac{2}{\gamma}\left(Su - 2Pv\right) \quad \Longrightarrow \quad \gamma v = \rho^0_{nn'} \frac{2\gamma^2 P}{2\left(S^2 - 4P^2\right) + \gamma^2 S}. \tag{13}$$

With $\Omega_n^2 = g^2 n/4$ one has $S = \frac{g^2}{4}(n+n')$, $P = \frac{g^2}{4}\sqrt{nn'}$ and $S^2 - 4P^2 = \frac{g^4}{16}(n-n')^2$, so all powers of g cancel except one ratio:

$$\gamma v = \rho^0_{nn'} K_{nn'}, \qquad K_{nn'} = \frac{2\sqrt{nn'}}{\dfrac{g^2}{2\gamma^2}\left(n-n'\right)^2 + n + n'}. \tag{14}$$

Note $K_{nn} = 1$: every $|b, n \geq 1\rangle$ ends up in $|d, n-1\rangle$ with probability one, as it must.

2.4 Result

Equation (14) says $\langle n-1|\hat{\rho}_{dd}(\infty)|n'-1\rangle = \rho^0_{nn'} K_{nn'}$. Relabelling $n \to n'+1$, $n' \to n+1$ and restoring the dark corner of Eq. (8):

Result — any input state

$$\boxed{\langle n'|\hat{\rho}^{\text{ss}}_c|n\rangle = \rho^0_{00}\,\delta_{n0}\delta_{n'0} + \rho^0_{n'+1,n+1}\,\frac{2\sqrt{(n+1)(n'+1)}}{\dfrac{g^2}{2\gamma^2}\left(n-n'\right)^2 + n + n' + 2}} \tag{15}$$

equivalently, in operator form with $\circ$ the elementwise (Hadamard) product in the Fock basis,

$$\hat{\rho}^{\text{ss}}_c = \rho^0_{00}|0\rangle\langle 0| + F \circ \left(\hat{a}\,\hat{\rho}_c(0)\,\hat{a}^\dagger\right), \qquad F_{n'n} = \frac{2}{\dfrac{g^2}{2\gamma^2}(n-n')^2 + n + n' + 2}. \tag{16}$$

For the coherent state of the problem, $\rho^0_{n'n} = \mathrm{e}^{-|\alpha|^2}\alpha^{n'}(\alpha^*)^n/\sqrt{n'!\,n!}$, and Eq. (15) becomes

$$\boxed{\langle n'|\hat\rho_c^{\rm ss}|n\rangle = \mathrm{e}^{-|\alpha|^2}\left[\delta_{n0}\delta_{n'0} + \frac{2\,|\alpha|^2\,\alpha^{n'}(\alpha^*)^n}{\sqrt{n!\,n'!}\,\left[\dfrac{g^2}{2\gamma^2}(n-n')^2 + n + n' + 2\right]}\right]} \tag{17}$$

where $\mathrm{e}^{-|\alpha|^2} \equiv \exp(-|\alpha|^2)$.

3 Physical content and checks

Trace. $\sum_n \langle n|\hat\rho_c^{\rm ss}|n\rangle = \mathrm{e}^{-|\alpha|^2}\left[1+\sum_n |\alpha|^{2(n+1)}/(n+1)!\right] = \mathrm{e}^{-|\alpha|^2}\mathrm{e}^{|\alpha|^2} = 1$.

The diagonal is trivial and g,γ-independent. Setting $n=n'$ in Eq. (15) gives $\langle n|\hat\rho_c^{\rm ss}|n\rangle = \rho^0_{n+1,n+1} + \rho^0_{00}\delta_{n0}$: exactly one photon is removed unless the cavity was already empty. All the physics of g and γ lives in the *coherences*.

The coherence-reduction factor. Dividing by $\sqrt{\rho^{\rm ss}_{nn}\rho^{\rm ss}_{n'n'}}$, the normalised coherence of the initially pure state is multiplied by

$$D_{n'n} = \frac{2\sqrt{(n+1)(n'+1)}}{\dfrac{g^2}{2\gamma^2}(n-n')^2 + n + n' + 2} \;\le\; 1, \qquad D_{nn} = 1, \tag{18}$$

by AM–GM. Two limits:

- $g/\gamma \to \infty$: $D \to 0$ for $n \neq n'$ — complete decoherence in the Fock basis. The emitted photon is a clock: the vacuum Rabi frequency $\Omega_n = \frac{g}{2}\sqrt{n}$ is resolved within the excited-state lifetime, so the emission time carries full which-path information about n.
- $g/\gamma \to 0$: $D \to 2\sqrt{(n+1)(n'+1)}/(n+n'+2) < 1$ — decoherence survives even in the Purcell limit, now because the *rates* $\Gamma_n = 4\Omega_n^2/\gamma = g^2 n/\gamma$ differ with n: an exponential decay with an n-dependent time constant still leaks which-path information.

For example $D_{01} = 2\sqrt{2}/(3+\frac{g^2}{2\gamma^2})$: 0.9412 at $g/\gamma = 0.1$, 0.5657 at $g/\gamma = 2$, 0.0534 at $g/\gamma = 10$.

4 QuTiP cross-check

`answer.py` propagates Eq. (1) with `qutip.mesolve` to $T = 60/\min(\gamma/4,\, g^2/\gamma)$ (the slowest relaxation rate of the $n=1$ manifold), traces out the atom, and compares with Eq. (15). The same checks are also available as the standalone script `qutip_check.py`, in which the analytic formula is re-implemented from scratch — so the agreement below is not an artefact of shared code — and which additionally shows the degeneracy of the Liouvillian kernel explicitly ($N^2+1+2N = 49$ zero eigenvalues at $N=6$, with `qutip.steadystate()` failing outright on the singular matrix), the exponential convergence in T, and a printed 5×5 corner of both matrices.

Two remarks make the comparison exact rather than approximate. First, a Fock cutoff at N is *not* an approximation here: $\hat H$ only converts $|b,n\rangle$ into $|e,n-1\rangle$, so the photon number never increases and $\mathrm{span}\{|n\rangle : n<N\}$ is invariant. Second, the same truncated and renormalised amplitude vector is fed to both sides, so the column "formula (15)" below is a pure test of the physics, limited only by the integrator tolerances (`atol` $=10^{-12}$, `rtol` $=10^{-10}$); the "closed form" column additionally carries the $N\to\infty$ truncation error of Eq. (17), listed separately.

Structural checks at $g=2, \gamma=1, \alpha=1.1, N=24$: $\mathrm{Tr}\,\hat\rho_c^{\rm ss} = 1.00000000000000$; $\max|\rho-\rho^\dagger| = 1.7\times 10^{-18}$; $\max_n\left|\rho^{\rm ss}_{nn} - (|c_{n+1}|^2 + |c_0|^2\delta_{n0})\right| = 2.8\times 10^{-17}$.

g	γ	α	N	$\max\lvert\rho^{(15)} - \rho^{\mathrm{QuTiP}}\rvert$	$\max\lvert\rho^{(17)} - \rho^{\mathrm{QuTiP}}\rvert$	truncation
1.0	1.0	1.0	20	1.36×10^{-11}	1.36×10^{-11}	1.15×10^{-11}
10.0	1.0	1.0	20	5.32×10^{-11}	5.32×10^{-11}	1.30×10^{-13}
0.2	1.0	1.2	22	2.50×10^{-13}	1.65×10^{-10}	1.65×10^{-10}
3.0	2.0	1.5	24	4.65×10^{-12}	1.01×10^{-10}	1.01×10^{-10}
2.0	0.7	$0.8 + 0.6\mathrm{i}$	22	3.13×10^{-11}	3.13×10^{-11}	6.21×10^{-14}

Table 1: Coherent input. The parameter sweep spans deep weak coupling ($g/\gamma = 0.2$) to deep strong coupling ($g/\gamma = 10$) and a complex α. Generated by `answer.py`.

5 Bonus: squeezed input

Nothing in Sec. 2 used the coherent-state amplitudes — Eq. (15) is linear in $\hat{\rho}_c(0)$ and holds for *any* input, pure or mixed. For a squeezed vacuum $|\zeta\rangle = \hat{S}(r\mathrm{e}^{\mathrm{i}\theta})|0\rangle$,

$$s_{2m} \equiv \langle 2m|\zeta\rangle = \frac{\left(-\mathrm{e}^{\mathrm{i}\theta}\tanh r\right)^m \sqrt{(2m)!}}{2^m\, m!\,\sqrt{\cosh r}}, \qquad s_{2m+1} = 0, \tag{19}$$

so that

$$\boxed{\langle n'|\hat{\rho}_c^{\mathrm{ss}}|n\rangle = |s_0|^2\delta_{n0}\delta_{n'0} + s_{n'+1}s_{n+1}^{*}\,\frac{2\sqrt{(n+1)(n'+1)}}{\dfrac{g^2}{2\gamma^2}(n-n')^2 + n + n' + 2}} \tag{20}$$

with the striking consequence that **the steady state is supported on *odd* photon numbers plus the vacuum**: the squeezed vacuum contains only even n, and exactly one photon is always removed. The surviving vacuum weight is $|s_0|^2 = 1/\cosh r$. Table 2 confirms Eq. (20), and the last two rows of Table 3 confirm the general formula (15) for a squeezed *coherent* input $\hat{D}(\alpha)\hat{S}(r)|0\rangle$, for which both parities are populated.

g	γ	r	θ	N	$\max\lvert\rho^{(20)} - \rho^{\mathrm{QuTiP}}\rvert$	$\mathrm{Tr}\,\rho$	$P(\mathrm{odd}) + P(0)$
1.0	1.0	0.6	0.0	24	2.07×10^{-11}	1.000000000000	1.0000000000
5.0	1.0	0.8	0.0	28	2.18×10^{-11}	1.000000000000	1.0000000000
0.3	1.0	0.5	0.0	24	2.12×10^{-12}	1.000000000000	1.0000000000
2.0	1.5	0.7	0.9	26	1.02×10^{-11}	1.000000000000	1.0000000000

Table 2: Squeezed-vacuum input. The last column verifies that all the population really does sit on odd n plus the vacuum.

g	γ	α	r	N	$\max\lvert\rho^{(15)} - \rho^{\mathrm{QuTiP}}\rvert$
1.5	1.0	0.7	0.4	24	3.53×10^{-12}
3.0	1.0	$0.5 + 0.3\mathrm{i}$	0.6	26	2.89×10^{-11}

Table 3: Squeezed coherent input $\hat{D}(\alpha)\hat{S}(r)|0\rangle$.

Worst disagreement over all eleven cases: 5.3×10^{-11}, consistent with the requested integrator tolerance.